

STANDARD OPERATING PROCEDURE: HPLC SYSTEM

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1.0 PURPOSE

To provide a standardized methodology for the startup, operation, and shutdown of the High-Performance Liquid Chromatography (HPLC) system to ensure reproducible analytical results and prevent column degradation.

2.0 SCOPE

This procedure applies to all laboratory personnel utilizing the HPLC system for the analysis of active pharmaceutical ingredients (APIs) and finished products.

3.0 SAFETY & PRECAUTIONS

- **PPE:** Wear chemical-resistant gloves, lab coat, and safety goggles at all times.
- **Pressure Limits:** Never exceed the maximum pressure rating of the column (typically 400 bar for analytical columns).
- **Solvent Compatibility:** Ensure the mobile phase is compatible with the column stationary phase to prevent irreversible bed collapse.
- **Degassing:** All solvents must be HPLC-grade and degassed for 15 minutes via sonication or inline vacuum degassing to prevent baseline noise.

4.0 PRE-OPERATIONAL CHECKS

1. Verify that solvent reservoirs have sufficient volume for the planned run.
2. Check the waste container level; ensure it is not more than 75% full.
3. Inspect all PEEK and stainless-steel tubing for leaks or salt crystallization at the fittings.

4. Ensure the column installed matches the specific "Method Validation" requirement for the sample being tested.

5.0 OPERATING PROCEDURE

5.1 System Priming and Equilibration

1. Open the purge valve on the pump.
2. Set the flow rate to 5.0 mL/min and flush each solvent line for 3 minutes to remove air bubbles.
3. Close the purge valve.
4. Set the analytical flow rate (typically 1.0 mL/min) and allow the system to pump the mobile phase through the column.
5. Monitor the pressure and the UV-Vis detector baseline. Equilibration is complete when the baseline drift is less than 0.5 mAU/hr and pressure fluctuation is <1%.

5.2 Sample Injection

1. Verify that samples have been filtered through a $0.45\mu\text{m}$ syringe filter to prevent frit clogging.
2. Load vials into the autosampler tray.
3. Input the "Sequence" into the software (Sample ID, Injection Volume, and Run Time).
4. Perform a "Blank Injection" (mobile phase only) first to ensure the system is free of carryover contamination.

6.0 SHUTDOWN AND CLEANING

1. **Column Wash:** After the final run, flush the column with a "Storage Solvent" (typically 50/50 Water/Methanol or Acetonitrile) for 30 minutes to remove buffered salts. Failure to remove salts will lead to pump seal damage.
2. Reduce flow rate gradually to 0.0 mL/min before turning off the pump.

3. Turn off the detector lamp to conserve its lifespan (typically 2,000 hours).

7.0 REVISION HISTORY

Version	Description of Change	Author	Date
1.0	Initial Release: Standardized HPLC Protocol	Coker Rebecca	07-Jan-2026